IIIF Document Information Extraction — Weekly Briefing

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Meeting: 11:00–11:20 · **Room:** INN 128 · **Zoom:** https://epfl.zoom.us/j/2693459226

Oral update

Good morning. This week, I studied the main IIIF discovery mechanisms and designed a seven-module harvesting pipeline. I implemented and tested the first three modules: provider census and selection, entry-URL validation, and IIIF resource discovery.

Module 1 currently combines 95 unique candidate records: 69 from the official IIIF Finding Resources guide, 25 from Awesome IIIF, and one from the IIIF Registry. Module 2 checks HTTP status, redirects, response type, elapsed time, JSON parsing, and basic IIIF type. Module 3 discovers resources through Change Discovery streams, IIIF Collections, direct Manifest links, and links declared in HTML pages.

The tested routes produced 21,872 unique Manifest URLs: 21,853 from the Bodleian Change Discovery stream, 18 from Fragmentarium, and one from a Biblissima demonstration. This is a reproducible baseline, not the total available from all 95 candidates.

Testing revealed that the catalogue converter normally retains only the first source-entry link, which is often a homepage. Institutions instead expose IIIF through Change Discovery, Collections, institutional APIs, URL templates, or record-level links. I therefore propose to retain and classify all relevant links in Module 1, validate every candidate endpoint in Module 2, and add Collection pagination plus API/template adapters in Module 3. I would like to confirm this architecture, the first complete provider, the rights allowlist, and the common-schema handoff.

Verified evidence

Area	Result
Module 1	95 candidates = 69 official-guide + 25 Awesome IIIF + 1 Registry + 0 manual
Module 2	1 reachable JSON; 65 reachable non-JSON; 1 too large; 12 restricted; 4 temporary failures; 12 unreachable
Module 3	21,872 unique Manifest URLs = Bodleian 21,853 + Fragmentarium 18 + Biblissima demo 1
Tests rerun today	Module 1: 13; Module 2: 11; Module 3: 16 — **40/40 passed**
Meaning	Candidates are not confirmed APIs; HTML or one failed URL does not prove that an institution lacks IIIF

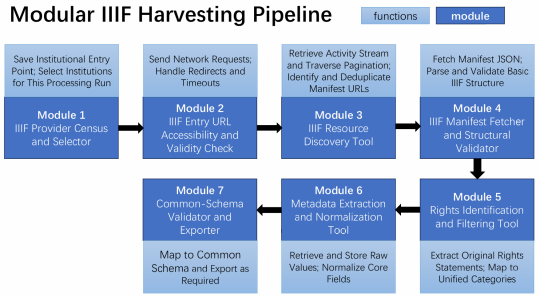


Figure 1. Seven-module harvesting pipeline

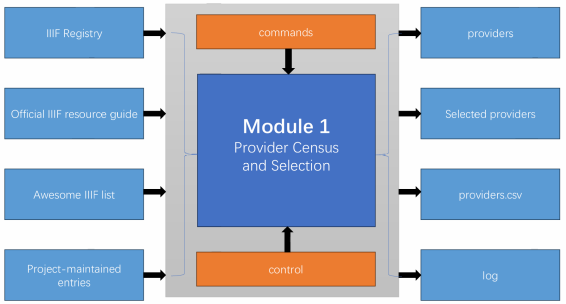


Figure 2. Module 1 inputs and outputs

Core problem

IIIF has no mandatory universal discovery endpoint. The Registry is voluntary and incomplete; the official guide mainly explains record-level Manifest discovery; Awesome IIIF mixes providers, viewers, software, Presentation API and Image API implementations. Module 1 also discarded secondary links, so present coverage measures input and adapter limitations rather than worldwide IIIF availability.

A reachable HTML page may still lead to IIIF. A 403/500 response may indicate restrictions, temporary failure, or a moved endpoint. An Image API **info.json** describes image delivery and is not a Presentation Manifest.

Required corrections

- **Module 1:** preserve the raw entry and every relevant link; classify homepage, Change Discovery, Collection, search/API documentation, Manifest examples and URL templates; distinguish providers from viewers, software and Image-only services.
- **Module 2:** check every structured candidate link and retain status, redirect, final URL, content type, timing, JSON/type evidence and failures; do not stop after one success.
- **Module 3:** add Presentation 2/3 Collection **next** pagination, API/template adapters, loop detection, limits, checkpoints, caching, rate limiting, retry/backoff and resumable output.

Use bounded samples before full harvesting. BSB advertised about 3.2 million Collection members; Carnegie exposed about 908 Manifests.

Essential interfaces

- **Registry:** locates voluntarily registered Change Discovery streams; not a complete census.
- **Change Discovery:** follow OrderedCollection/Page; read orderedItems and object.id; exclude deleted/removed objects.
- **Collection:** Presentation 2 uses manifests/collections; Presentation 3 uses items; follow next with loop detection.
- **Manifest:** describes one digital object; Presentation 2 commonly uses sequences, Presentation 3 uses items.
- **Image API:** provides info.json and image derivatives; it is not itself a Manifest.
- **Institutional API/template:** enumerate identifiers, then retrieve or construct Manifest URLs.

Command reference

```
# Generic JSON
curl.exe -L -H "Accept: application/ld+json, application/json" URL -o out.json
python -m json.tool out.json > pretty.json

# Registry and Change Discovery
curl.exe -L "https://registry.iiif.io/index.json" -o registry.json
curl.exe -L "https://iiif.bodleian.ox.ac.uk/iiif/activity/all-changes" -o activity.json

# Top Collections and one Manifest
curl.exe -L "https://api.digitale-sammlungen.de/iiif/presentation/v2/collection/top" -o bsb.json
curl.exe -L "https://cmoa-records-images.s3.amazonaws.com/collection/top.json" -o cmoa.json
curl.exe -L "https://fragmentarium.ms/metadata/iiif/F-nk2n/manifest.json" -o manifest.json
```

Other routes: Europeana Search/Record API plus keyed Presentation API; Harvard keyed Object API plus **/manifests/object/{OBJECT_ID}**; Gallica API plus **/iiif/ark:/12148/{ARK}/manifest.json**; Internet Archive search plus **/iiif/{ITEM_ID}/manifest.json**; Austrian National Library SACHA **/presentation/{projectName}/{id}/manifest**. Keep credentials outside Git.

Confirm with supervisors

1. Is the revised multi-strategy discovery architecture correct?
2. Which provider should be the first complete harvest?
3. What is the approved rights allowlist?
4. How should Image-API-only records be treated?
5. What are the common schema, handoff format, server resources and rate limits?

Immediate next actions

1. Fix Module 1 multi-link ingestion and add regression tests.
2. Extend Module 2 to validate all candidate links.
3. Add Collection pagination and API/template adapters to Module 3.
4. Run bounded BSB and Carnegie samples.
5. Implement Module 4 Manifest retrieval/validation, then rights filtering, metadata normalization and common-schema export.